UNIT–I: Data Communication & Switching

1. Define Data Communication. What are its basic components?

Data communication is the process of exchanging data between two or more devices through a communication channel.

The five fundamental components are:

1. **Message** – Information being communicated.
2. **Sender** – Device that originates the data.
3. **Receiver** – Device that receives the data.
4. **Transmission Medium** – Physical or wireless path used for transmission.
5. **Protocol** – Set of rules governing communication.

An effective communication system ensures **delivery, accuracy, timeliness, and appropriate jitter performance**.

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2. Differentiate between Analog and Digital Signals.

An **analog signal** varies continuously with time and can assume infinitely many amplitude values. A **digital signal** represents information using discrete signal levels, commonly binary **0 and 1**.

Parameter	Analog Signal	Digital Signal
Nature	Continuous	Discrete
Number of values	Infinite	Finite
Noise immunity	Lower	Higher
Regeneration	Difficult	Easier
Representation	Continuous waveform	Discrete signal levels

Digital signals are generally preferred in computer networks because they support efficient **regeneration and error-control mechanisms**.

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3. What is a Modem? Explain its function.

A **modem** is a communication device that performs:

$$\text{Modem} = \text{Modulator} + \text{Demodulator}$$

Modulation

Converts digital information into a signal suitable for the transmission medium.

Demodulation

Converts the received modulated signal back into digital information.

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Digital Data    Modulator    Channel    Demodulator    Digital Data
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Thus, a modem enables communication between digital systems across appropriate communication channels.

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4. Compare Circuit Switching, Message Switching, and Packet Switching.

Feature	Circuit Switching	Message Switching	Packet Switching
Path	Dedicated	Not dedicated	Not dedicated
Transmission unit	Continuous stream	Complete message	Packet
Connection setup	Required	Not required	Generally not required
Intermediate storage	Minimal	Entire message	Individual packets
Resource allocation	Reserved	Shared	Shared
Example	Traditional telephone network	Store-and-forward systems	Internet

Circuit switching reserves a complete path, **message switching** forwards the entire message using store-and-forward, and **packet switching** divides data into smaller packets for efficient transmission.

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5. Compare the OSI Model and TCP/IP Model.

Parameter	OSI Model	TCP/IP Model
Number of layers	7	4
Nature	Reference model	Protocol suite
Upper layers	Application, Presentation, Session	Combined as Application
Lower layers	Physical and Data Link separate	Combined as Network Access
Network layer	Network	Internet
Usage	Conceptual reference	Practical Internet architecture

The **OSI model** provides a structured framework for understanding network communication, whereas the **TCP/IP model** forms the basis of modern Internet communication.

UNIT–II: Data Link Layer

6. What is CRC? Explain its working principle.

Cyclic Redundancy Check (CRC) is an error-detection technique based on **modulo-2 division**.

Working:

1. A predefined **generator polynomial** is selected.
2. The sender appends zeros to the data.
3. Modulo-2 division is performed.
4. The resulting remainder is appended to the original data.
5. The receiver divides the received codeword by the same generator.

If:

$$\text{Remainder} = 0$$

the frame is considered free from detectable errors; otherwise, an error is detected.

CRC is particularly effective in detecting **burst errors**.

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7. What is Hamming Code?

Hamming Code is an error-control technique that uses redundant parity bits to detect and correct errors, particularly **single-bit errors**.

The parity bits are placed at positions that are powers of two:

$$1, 2, 4, 8, 16, \ldots$$

For `m` data bits and `r` redundant bits:

$$2^{\mathrm{r}} \geq \mathrm{m} + \mathrm{r} + 1$$

At the receiver, parity checks produce a **syndrome**, which identifies the position of an erroneous bit.

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8. Differentiate between Stop-and-Wait and Sliding Window Protocols.

Parameter	Stop-and-Wait	Sliding Window
Frames in transit	One	Multiple
Acknowledgment	Required before next frame	Multiple frames can be acknowledged
Throughput	Low	High
Complexity	Low	Higher
Buffer requirement	Minimal	Higher
Efficiency	Lower	Better

The **Sliding Window Protocol** improves channel utilization by permitting multiple outstanding frames.

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9. Differentiate between Go-Back-N ARQ and Selective Repeat ARQ.

Parameter	Go-Back-N	Selective Repeat
Retransmission	Lost frame and subsequent frames	Only lost/damaged frame
Receiver behavior	Discards out-of-order frames	Buffers out-of-order frames
Acknowledgment	Usually cumulative	Individual/selective
Buffer requirement	Lower	Higher
Complexity	Moderate	Higher
Efficiency	Lower during errors	Higher

Selective Repeat ARQ provides better bandwidth utilization, whereas **Go-Back-N ARQ** is simpler to implement.

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10. What is CSMA/CD and Binary Exponential Backoff?

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the classical medium-access mechanism used in shared, half-duplex Ethernet.

Basic operation:

1. Station senses the medium.
2. If idle, transmission begins.
3. The station monitors for a collision.
4. If a collision occurs, transmission stops.
5. A random backoff period is selected before retransmission.

The **Binary Exponential Backoff (BEB)** algorithm selects a random value:

$$\mathrm{r} \in [0, 2^{\mathrm{k}} - 1]$$

where `k` represents the number of collisions.

The waiting time is:

$$\text{Backoff Time} = \mathrm{r} \times \text{Slot Time}$$

The contention window increases after repeated collisions, reducing the probability of further collisions.

UNIT–III: Network Layer

11. What is the difference between Virtual Circuit and Datagram Networks?

Feature	Virtual Circuit	Datagram Network
Connection setup	Required	Not required
Route	Pre-established	May vary for each packet
Packet order	Generally maintained	May arrive out of order
Router state	Maintains connection state	Minimal per-flow state
Example concept	Connection-oriented network	IP-based Internet

A **Virtual Circuit** establishes a logical path before transmission, while a **Datagram Network** forwards each packet independently.

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12. What is Distance Vector Routing?

Distance Vector Routing is a dynamic routing technique in which each router maintains:

- The **distance (metric)** to each destination.
- The **next-hop router** used to reach that destination.

Routers periodically exchange their routing information with immediate neighbors.

It is based on the **Bellman-Ford algorithm**:

$$D_x(y) = \min_{v}\{c(x, v) + D_v(y)\}$$

Where:

- (D_x(y)) is the distance from router `x` to destination `y`.
- (c(x,v)) is the cost to neighboring router `v`.
- (D_v(y)) is the distance reported by neighbor `v`.

RIP is a well-known Distance Vector Routing Protocol.

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13. What is Link State Routing?

Link State Routing is a routing technique in which every router builds a database describing the network topology and independently computes the shortest paths.

The general process is:

1. Discover neighboring routers.
2. Determine link costs.
3. Generate Link State Advertisements (LSAs).
4. Flood LSAs throughout the network.
5. Build a complete topology database.
6. Apply the **Shortest Path First (Dijkstra) algorithm**.

Compared with Distance Vector Routing, Link State Routing generally provides **faster convergence and better scalability**, but requires greater processing and memory resources.

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14. What is OSPF?

Open Shortest Path First (OSPF) is a **link-state interior gateway routing protocol** used within an autonomous system.

Its major characteristics include:

- Uses the **Dijkstra Shortest Path First (SPF) algorithm**.
- Uses a **cost-based routing metric**.
- Maintains a **Link-State Database (LSDB)**.
- Supports hierarchical routing using **areas**.
- Uses flooding to distribute link-state information.
- Provides relatively fast convergence.

OSPF determines the lowest-cost path based on accumulated link costs.

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15. What is BGP?

Border Gateway Protocol (BGP) is an **inter-domain routing protocol** used for routing between **Autonomous Systems (ASs)**.

Unlike Distance Vector or Link State protocols, BGP is commonly classified as a **Path Vector Protocol**.

Major characteristics:

- Used between autonomous systems.

- Exchanges routing and reachability information.
- Uses policy-based route selection.
- Includes the **AS-PATH** attribute.
- Helps prevent inter-domain routing loops.
- Forms the routing infrastructure of the global Internet.

The AS-PATH records the autonomous systems through which a route has passed.
